

FREMONT, MI, USA

WMO: 720415

Lat: 43.433N

Lon: 86.000W

Elev: 831

StdP: 14.26

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 00140

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.6	5.6	-8.4	3.6	6.7	-3.5	4.7	11.8	24.7	29.0	21.0	24.6	4.3	270	0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	88.0	71.3	85.3	70.1	82.3	68.3	74.9	82.5	73.3	80.9	71.9	78.3	8.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.7	125.2	78.0	71.6	120.3	76.4	69.9	113.4	75.1	38.9	82.8	37.5	80.4	36.1	77.4	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.5	18.8	16.8	DB	-7.5	92.3	6.8	3.1	-12.4	94.5	-16.4	96.3	-20.2	98.1	-25.1	100.4	(4)
			WB	-8.0	77.9	6.8	1.9	-12.9	79.2	-16.9	80.3	-20.7	81.4	-25.7	82.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.6	22.8	24.6	33.8	45.1	58.0	66.5	70.6	68.6	62.3	49.9	38.5	29.5
	DBStd	18.78	10.02	10.22	10.53	8.76	8.77	6.03	5.89	5.25	7.31	8.21	8.63	8.56
	HDD50	3395	843	711	516	197	24	0	0	0	3	107	358	636
	HDD65	6859	1308	1131	969	598	253	52	18	25	138	471	795	1101
	CDD50	2536	0	0	12	50	271	497	638	578	371	105	14	0
	CDD65	523	0	0	0	1	35	98	191	138	56	4	0	0
	CDH74	4483	0	0	8	19	368	825	1763	1059	406	35	0	0
	CDH80	1156	0	0	1	0	91	196	548	227	91	2	0	0
Wind	WSAvg	6.7	7.5	7.8	7.5	8.6	7.0	6.2	5.1	4.7	5.3	6.6	7.3	7.4
Precipitation	PrecAvg	35.0	2.3	1.9	2.4	3.2	4.0	3.6	3.1	3.1	3.0	3.4	3.3	2.3
	PrecMax	43.3	3.9	3.9	5.1	5.2	9.9	6.0	7.3	5.6	7.1	7.7	7.2	5.4
	PrecMin	29.2	0.8	0.8	0.5	0.9	0.7	1.5	1.3	0.9	0.2	0.2	1.0	1.1
	PrecStd	4.4	1.1	0.8	1.3	1.2	2.2	1.4	1.4	1.4	1.6	2.0	1.9	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.5	55.9	75.0	77.0	87.8	89.4	92.3	88.3	88.2	79.1	67.5	57.0
		MCWB	46.8	49.5	63.5	58.1	69.5	74.4	74.4	71.6	69.3	63.8	55.4	53.9
	2%	DB	43.1	48.6	61.3	71.6	82.0	85.2	88.5	85.5	82.2	72.6	61.1	48.4
		MCWB	39.9	44.1	54.3	54.1	66.4	68.4	71.9	70.8	69.2	61.9	52.6	45.9
	5%	DB	38.4	41.7	55.0	65.3	77.8	81.8	85.8	82.3	78.8	68.1	55.2	44.5
		MCWB	36.2	37.6	47.6	51.4	62.1	66.8	71.1	69.0	65.9	60.5	49.1	42.2
	10%	DB	35.8	37.3	49.5	60.4	73.7	79.0	82.4	80.3	75.3	63.8	51.7	40.5
		MCWB	34.1	34.1	42.8	49.1	60.9	65.3	68.7	67.8	64.3	57.8	46.5	38.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.8	51.6	63.6	63.4	72.3	76.4	77.7	75.4	74.7	68.8	59.0	55.7
		MCDB	49.8	52.3	74.1	70.9	83.1	85.3	86.4	81.8	81.7	72.5	63.5	56.4
	2%	WB	40.3	44.2	57.1	57.7	69.1	72.4	75.6	74.0	72.1	65.5	54.0	46.4
		MCDB	42.1	49.5	59.6	64.6	77.5	80.4	83.6	80.6	78.1	68.6	59.1	47.9
	5%	WB	36.7	38.4	48.3	54.6	66.4	70.4	73.7	72.5	70.1	62.2	50.2	42.4
		MCDB	38.5	41.7	53.4	62.3	74.4	77.1	81.9	78.5	75.2	66.8	55.4	43.9
	10%	WB	34.2	34.8	43.1	50.6	63.6	68.5	72.2	70.9	68.1	58.4	46.5	38.0
		MCDB	35.5	37.3	48.5	58.3	70.3	75.0	79.2	76.1	73.3	63.3	50.3	40.1
Mean Daily Temperature Range	5% DB	MDBR	11.9	13.7	16.4	19.2	21.0	20.0	20.8	19.9	19.6	16.8	12.9	10.0
		MCDBR	13.9	17.3	23.0	27.2	25.3	23.5	23.2	22.1	22.6	23.0	19.4	15.7
		MCWBR	12.1	13.3	14.5	13.9	11.7	10.3	9.6	9.5	10.4	13.2	12.8	13.8
	5% WB	MCDBR	13.5	16.3	20.3	22.2	21.0	18.5	19.7	17.8	17.5	19.0	17.4	15.3
		MCWBR	12.2	14.0	14.7	15.1	11.5	10.4	9.8	9.3	10.0	14.3	13.5	14.2
Clear-Sky Solar Irradiance	taub	0.293	0.319	0.338	0.366	0.398	0.414	0.410	0.406	0.381	0.346	0.313	0.288	
	taud	2.369	2.289	2.363	2.318	2.286	2.288	2.298	2.313	2.361	2.454	2.487	2.437	
	Ebn at Noon	266	276	286	285	277	272	272	269	268	264	258	258	
	Edh at Noon	25	32	34	39	41	41	40	38	34	27	23	22	
All-Sky Solar Radiation	RadAvg	386	657	1132	1487	1757	1960	1992	1692	1348	828	482	311	
	RadStd	40	77	119	140	165	124	100	103	118	101	68	43	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page